LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:14

Aim

To perform image transformation using a Homography Matrix in Python with OpenCV and observe the transformed image.

Theory

A Homography matrix is a 3 × 3 matrix used to map points from one image plane to another.

It is commonly used for:

- Perspective transformation
- Image alignment
- Document scanning
- Image stitching
- Camera view transformation
- For four corresponding points, OpenCV can calculate the homography matrix using cv2.findHomography().

Algorithm

1. Import cv2 and numpy.
2. Read the input image.
3. Define four source points from the image.
4. Define four destination points.
5. Calculate the Homography matrix using cv2.findHomography().
6. Apply the Homography matrix using cv2.warpPerspective().
7. Display the original and transformed images.

8. Press q to close the windows.

CODE:

EXP-14.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Read the image
image = cv2.imread("krishna.jpeg")

if image is None:
    print("Error: Image not found!")
    exit()

# Get image dimensions
height, width = image.shape[:2]

# Source points
src_points = np.float32([
    [100, 100],
    [width - 100, 100],
    [width - 100, height - 100],
    [100, height - 100]
])

# Destination points
dst_points = np.float32([
    [0, 0],
    [width, 0],
    [width, height],
    [0, height]
])

# Calculate Homography Matrix
H, status = cv2.findHomography(src_points, dst_points)

print("Homography Matrix:")
print(H)

# Apply Homography Transformation
transformed = cv2.warpPerspective(
    image, H, (width, height)
)
```

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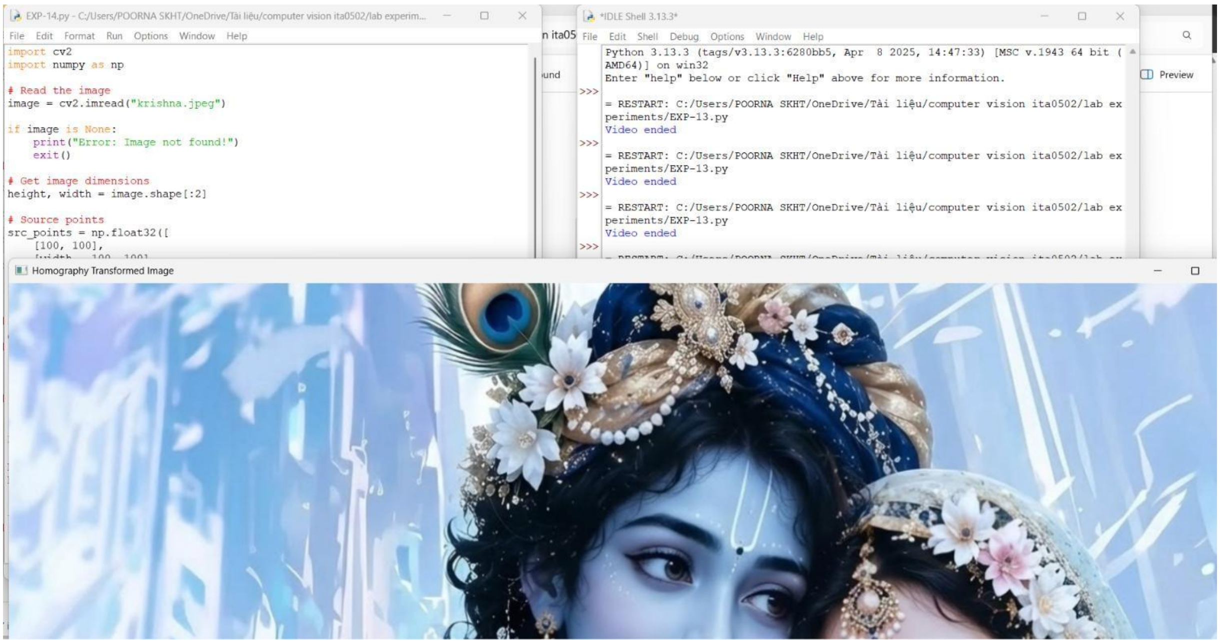
OUTPUT :



Result:

The Homography Matrix was successfully calculated using four corresponding points, and the image was transformed using cv2.warpPerspective(). The perspective of the selected region was changed successfully**.**